

Research Proposal

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Date: 01.05.2026

Preliminary title:

Coercion and Vulnerability: District-Level Land Use Consolidation under Rwanda's Crop Intensification Program and Smallholder Resilience to Agricultural and Weather Shocks.

Research question:

How does district-level intensity of land use consolidation under Rwanda's Crop Intensification Program shape the likelihood that households resort to distress coping strategies following an agricultural or weather shock, and does this effect vary by household wealth?

Sub-questions:

- Is district-level LUC intensity associated with a higher likelihood of distress coping among households that experienced an agricultural or weather shock?
- Does this effect increase as household wealth decreases?

Current knowledge:

Rwanda's Crop Intensification Program (CIP), launched in 2007 and ongoing, mandates smallholder farmers to cultivate state-assigned monocrops on consolidated plots, enforced through the *imihigo* performance contract system (Cioffo et al., 2016). The programme is widely cited as a success, producing nationally reported yield increases—though these have been contested on data quality grounds (Heinen, 2022). Existing research reveals a distributional cost: Del Prete et al. (2019), using EICV panel data from 2010/11 and 2013/14, find that land consolidation participation had ambiguous effects on dietary diversity, with no statistically significant overall DDS effect, but significant declines in consumption of meat, fish, and fruit. Crucially, they do not test whether these costs intensify when households face adverse shocks. Clay and Zimmerer (2020) argue qualitatively that the CIP eroded the crop diversity and livelihood flexibility that previously buffered smallholders against shocks, and that this vulnerability is unevenly distributed across wealth groups. Their claim has not been tested quantitatively. This study addresses that gap using more recent nationally representative data.

The core argument is that *imihigo* enforcement removed the micro-level adaptive strategies smallholders use to manage risk—intercropping, crop diversification across moisture gradients, and flexible plot allocation—creating structural vulnerability (Clay and Zimmerer, 2020; Matin et al., 2018). CIP households locked into monocrops of government-selected crops, which Clay and Zimmerer show are among the riskiest in drought and heavy rain, have fewer options to absorb losses when shocks hit. This cost falls hardest on the poorest households, who additionally lack savings, credit access, and non-farm income as alternative buffers (Matin et al., 2018).

Hypotheses:

H1 Among households that experienced an agricultural or weather shock, those in higher-LUC

districts will be more likely to resort to distress coping strategies than those in lower-LUC districts (*resilience-reducing effect*).

H2 This effect will be larger for poorer households (*poverty-concentration hypothesis*).

Preliminary structure:

1. *From Policy Success to Distributional Question* — Introduction and relevance
2. *What the Literature Misses* — State of the art and the untested resilience claim
3. *Coercion, Specialisation, and Structural Vulnerability* — Theory and hypotheses
4. *Measuring the Cost* — Research design and operationalization
 - 4.1 Data and variables
 - 4.2 Methodology and results
5. *Who Bears the Burden?* — Conclusion and implications

Research design:

The study matches district-level LUC intensity—constructed as a population-weighted average share of agricultural land under consolidation from the NISR Seasonal Agricultural Survey (SAS) 2024 across all three seasons—to household-level data from the EICV7 (2023/24) survey. The analysis is restricted to households that reported experiencing an agricultural or weather shock in the past 12 months (heavy rain, floods, landslides, drought, crop pests or disease, livestock disease, or unusually high agricultural input prices), identified from the EICV7 household shock module ($n \approx 3,000$).

The dependent variable is a binary indicator of distress coping: whether the household’s primary coping response was harmful or irreversible—reduced food expenditure, sold land or productive assets, sold last female animals, withdrew a child from school, or resorted to begging—as opposed to adaptive strategies such as using savings, borrowing from a bank, or working additional hours. The analysis uses logistic regression with LUC intensity as the main independent variable, a $LUC \times$ wealth-quintile interaction term to test H2, and province fixed effects and urban/rural controls throughout.

Data sources:

Variable	Source	Access
Distress coping indicator	EICV7 (2023/24) household module (s5eq3a)	World Bank Micro-data Catalog; free registration
Shock type & indicator	EICV7 (2023/24) household module (s5eq1, s5eq2a)	World Bank Micro-data Catalog; free registration
Household controls	EICV7 poverty file; savings & credit modules	Included in EICV7 package
LUC intensity	NISR Seasonal Agricultural Survey 2024, Seasons A, B & C	NISR microdata catalog; free registration

Literature:

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- Cioffo, Giuseppe, An Ansoms, and Jude Murison. 2016. “Modernising Agriculture through a ‘New’ Green Revolution.” *Review of African Political Economy* 43(148): 188–204.
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- Heinen, Sebastian. 2022. “Rwanda’s Agricultural Transformation Revisited.” *Journal of Development Studies* 58(10): 2044–2064.
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- Nsabimana, Alain, et al. 2023. “The Short and Long Run Effects of Land Use Consolidation Programme on Farm Input Uptakes.” *Land Use Policy* 132: 106787.